

- 1** The most common oxidation state of americium, Am, in aqueous solution is +3.

Recently, Cu^{3+} has been shown to quantitatively oxidise $\text{Am}^{3+}(\text{aq})$ in dilute HNO_3 , while itself is reduced to Cu^{2+} .

In an experiment, 20.0 cm^3 of $0.0120 \text{ mol dm}^{-3} \text{ Am}^{3+}(\text{aq})$ was found to require 24.00 cm^3 of $0.0300 \text{ mol dm}^{-3} \text{ Cu}^{3+}$ for complete oxidation.

What is the formula of the americium-containing species formed?

- A** $\text{Am}_2\text{O}_2^{2+}$
- B** AmO_2^{2+}
- C** AmO^{2+}
- D** AmO^{+}